This is the authors' version of the following article: Lee, H., Lee, J. H., & Scarcella, R. C. (2026). Influencing language mindsets and English writing competence through an EAP program: A longitudinal study with latent transition analysis. *Language Teaching Research, 30*(1), 92–114. The version of record is available at https://doi.org/10.1177/13621688221143258. This version differs from the published article in layout and pagination; please cite the published version.

Influencing language mindsets and English writing competence through an EAP program: A longitudinal study with latent transition analysis

Hansol Lee[a], Jang Ho Lee[b,*], Robin C. Scarcella[c]

[a] Korea Military Academy, South Korea

[b] Chung-Ang University, South Korea

[c] University of California, Irvine, USA

[*] Corresponding author: Jang Ho Lee, Department of English Education, Chung-Ang University, Building 203-427, 84 Heukseok-ro, Dongjak-gu, Seoul, South Korea. Email: jangholee@cau.ac.kr

Abstract

This study examined changes in undergraduate students' language mindsets and academic writing competence as a result of taking a one-semester English for academic purposes (EAP) writing course with a pre–post intervention on mindset. A total of 319 students participated in the study and completed a 10-week EAP course. The participants were administered the mindset questionnaire and the writing test at two different time-points. The results showed that the participants' level in one component of growth language mindset (general language intelligence beliefs), and scores on the writing test increased significantly. The results of the latent profile analysis revealed three distinctive profiles among the participants at both time-points: Growth profile, Mixed profile, and Fixed profile. The Growth and Mixed profiles were found to perform significantly better than the Fixed profile on the writing tests. Finally, the result of a latent transition analysis revealed that dynamic shifts in the profile membership could occur through the mindset intervention.

Keywords: EAP, language beliefs, language mindsets, latent transition analysis, person-centered approach

I Introduction

Mindset is an important subject of academic inquiry, as a psychological construct that represents one's beliefs about whether one's intellectual abilities and personality are improvable (Dweck, 1999; Yeager et al., 2019). This construct has been proposed to account

for people's goal-orientation, self-regulation, and reaction to challenges, as well as achievements (or learning outcomes) (Blackwell et al., 2007; Dweck & Leggett, 1988; Dweck & Yeager, 2019; Hong et al., 1999; Yeager et al., 2019). Mindsets can be distinguished into the growth mindset, under which intellectual abilities and other characteristics are believed to be subject to improvement, or the fixed mindset, under which they are believed to be immutable (Dweck, 2006).

With the notion that mindset may be domain-specific (Dweck, 2006; Dweck et al., 1995; in the second language learning literature, see Khajavy et al., 2022 and Waller & Papi, 2017), the concept has recently attracted some attention in second language (L2) learning literature, with Lou and Noels' (2017, 2019a, 2019b, 2020) Language Mindsets Inventory (LMI) as the major framework. Lou and Noels (2017, 2019a, 2019c) proposed that language mindsets could account for language learners' motivation and goal orientations, and may work in complex manners in concert with other variables related to language learning, such as course engagement, persistence, and adaptation to learning situations (Lou et al., 2022), making it a psychological concept worth examining.

Although the field has seen a recent surge of empirical studies on language mindsets (Eren & Rakıcıoğlu-Söylemez, 2020; Khajavy et al., 2021, 2022; Lou et al., 2022; Wang et al., 2021), most of these studies are cross-sectional. Scant research has implemented mindset intervention and examined whether learners' mindsets could be influenced by such interventions and how mindsets are associated with language learning over time (Lou et al., 2022). Empirical evidence obtained from such research could significantly advance our understanding of how language mindsets work in the long term in the context of L2 learning.

Employing the research design of the pre–post intervention, the present study was conducted to fill this gap in the L2 learning field, by examining changes in undergraduate students' language mindsets and academic writing competence as a result of completing an English for academic purposes (EAP) writing course at a large public university on the West Coast of the U.S. This writing course was deliberately equipped with the mindset intervention based on Dweck's (2006) influential work, with the goal to positively change the target students' mindsets. In addition to examining the whole group's changes in language mindsets and writing competence (i.e. variable-centered analysis), the present study further employed a person-centered analysis approach (Howard & Hoffman, 2018; Lee et al., 2019; von Eye & Bogat, 2006) to unearth learner profiles with regard to the aforementioned variables in view of recent studies that pointed to the value of such an approach to the topic (Lou et al., 2022; Yao, Wang, et al., 2021), and explore the shifts in the profile membership, if any, over the period of one semester.

II Background

1 The impact of academic writing programs

EAP programs are geared toward enhancing students' abilities to comprehend and use English in academic contexts (Hyland, 2016). Although these courses are designed to cover several linguistic dimensions (e.g. vocabulary, listening, spoken interaction), writing is considered an essential component, and is 'often the only means by which students' content knowledge is assessed in a large number of disciplines' (Mazgutova & Kormos, 2015, p. 3). In these courses, particularly those oriented toward developing one's academic writing competence, students are expected to read academic texts on a wide range of topics including those related to their own major, write (multiple) drafts on what they have read, and receive written feedback on their drafts from their instructors (Bitchener & Ferris, 2012; Ferris, 2002; Leki, 2006). Such processes are often exercised within task-based (Skehan, 1996) and/or genre-based approaches (Hyland, 2007, 2008).

The extent to which these EAP programs could improve students' academic writing competence has been systematically analysed in Lee and Lee's (2022) meta-analysis of 14 studies (175 effect sizes). It revealed that EAP programs could improve students' writing competence, with an average effect size (ES) of d = .45, although a range of moderator variables may influence the impact of EAP programs in complex manners. Moreover, this meta-analysis proposed domain-specific ES benchmarks based on the results of the identified studies, in which 0.3, 0.5, and 0.7 are considered small, medium, and large ES, respectively. In addition, it calls for further studies that address various course features of EAP programs, which may be related to the type of instruction, curriculum, or course materials. In this study, we answer this call by examining the development of English writing competence through an EAP course equipped with theories of mindset as set out in Dweck (1999, 2006). This is motivated by the findings that endorsing a growth mindset among L2 learners may help sustain their writing motivation (e.g. Waller & Papi, 2017; Yao, Guo, et al., 2021).

2 The role of language mindsets in language learning

Building on the studies on mindsets (Dweck, 1999, 2006; Dweck & Leggett, 1988) as well as those on language learners' beliefs about the role of intellectual abilities (Horwitz, 1988; Mercer & Ryan, 2010; Ryan & Mercer, 2012), Lou and Noels (2017, 2019a, 2019b) proposed a domain-specific scale of mindsets for language learning contexts. According to Lou and Noels, language mindsets cover three different but inter-related themes (general language intelligence beliefs, L2 aptitude beliefs, and age-sensitivity beliefs about language learning), for each of which learners may have different degrees of growth and fixed mindsets. Lou and

Noels (2019a) note that these three themes constitute not only scholarly issues in the second language acquisition (SLA) field, but also casual ideas about L2 learning in the public domain, as people try to make sense of their own or others' L2 learning experience in reference to ideas circulating around these beliefs.

These three themes have different theoretical backgrounds. First, general language intelligence beliefs (GLB), which are conceptually similar to linguistic intelligence in multiple intelligence theory (Gardner, 1993), are concerned with whether general language intelligence is mutable (or not). Second, L2 aptitude beliefs (L2B) correspond to the issue of whether one's ability to learn an L2 is fixed, with early SLA studies having addressed the role of L2 aptitude in L2 learning (see Carroll, 1981). Third, age-sensitivity beliefs about language learning (ASB) concern one's belief about whether there exists a critical/sensitive period after which one cannot achieve a native-like proficiency in the L2 (Birdsong, 1999; DeKeyser, 2000).

With these themes in mind, in their seminal study (2017), Lou and Noels developed the LMI, which consists of 18 questionnaire items (nine for growth and nine for fixed language mindsets across three themes), with 1,633 undergraduate students registered in a psychology course at a Canadian university. Confirmatory factor analysis found that their data fitted the two-factor second-order hierarchical model, with the three first-order fixed factors (GLB, L2B, and ASB) being loaded on a second-order fixed factor, and the three first-order growth factors being loaded on a second-order growth factor.

With the idea that mindsets are subject to change via interventions (Blackwell et al., 2007), Lou and Noels' (2016) study based on a randomized controlled trial with 150 university students revealed that learners who read a mock article promoting growth language mindsets react more positively to failure situations with a greater effort and intention to continue their L2 learning than their peers who read an article promoting fixed language mindsets. This finding corresponds to those of earlier studies in other domains that learners' mindsets are related to their responses to failure, and those with growth mindsets being more determined to overcome difficult situations with effort (e.g. Blackwell et al., 2007; Hong et al., 1999).

3 A person-centered approach to the language learning research

Although a variable-centered approach has taken precedence in quantitative methods in language learning research, a person-centered approach has gained attention from the research community as a complement to or alternative for the former (Howard & Hoffman, 2018; Lee et al., 2019; von Eye & Bogat, 2006). One aim of the person-centered approach is to unearth group memberships (Crowther et al., 2021; Staples & Biber, 2015), which is not the focus of the variable-centered approach (e.g. regression analysis).

Among those that have harnessed a person-centered approach in language learning research, one branch has been dedicated to identifying learner profiles in terms of participants' beliefs about (or attitude toward) learning-related variables and their language learning achievements. For example, by using a cluster analysis technique, Yamamori et al. (2003) were able to identify four clusters of learners among their sampled participants (secondary-level Japanese learners of English as a foreign language), who distinctively differed in their desire to learn English, language-learning strategies, and English achievements. Similarly, Matsumoto et al. (2013) identified different profiles among undergraduate Japanese learners of English as a foreign language (EFL), who received reading strategy intervention. With a longitudinal design, Matsumoto et al. were able to demonstrate how the identified clusters showed different learning trajectories (before and after the strategy intervention).

Language mindsets – the topic of the present study – appear to align well with the rationale of the aforementioned person-centered approach, in view of previous findings (Mercer & Ryan, 2010) that some language learners may have a mix of fixed and growth mindsets, and others may be oriented toward either fixed or growth mindsets.

Inspired by previous studies on mindsets based on a person-centered approach in other domains such as science and math (e.g. Chen & Tutwiler, 2017; Yu & McLellan, 2020), Lou et al. (2022) examined the mindset system profiles of L2 learners by employing a latent profile analysis (LPA), which is one of the statistical methods adopted in the person-centered approach. With 234 freshmen registered in foreign language courses in a Canadian university, they found that more than half the participants (about 57.7%) had a mixed profile, whereas the rest were oriented toward either growth (20.5%) or fixed (21.8%) profiles. Their results further revealed that different language mindset profiles showed different levels of the variables including mastery goals, persistence, and classroom and language anxiety. However, Lou et al. (2022) acknowledged that their study was cross-sectional, and suggested that future research examine the changes in language mindsets over an extended period of time in concert with its dynamics with language learning.

Another recent L2 work on mindset utilizing a person-centered approach is by Yao, Wang, et al. (2021), in which the changes in motivation and mindset related to English writing among undergraduate Chinese EFL learners were examined over 15 weeks. In this study, both experimental and control groups were given three English writing assignments; however, only the experimental group was given an opportunity to discuss their writing with their peers and received comments from an automatic writing evaluation (AWE) program. Their results revealed that the experimental group demonstrated higher levels of mindset and motivation after the intervention. The results of the person-centered analyses (i.e. a latent profile analysis

and a latent transition analysis) pointed to the distinct profiles related to the language mindset, as well as some dynamic changes in the profile membership among the participants as a result of the 15-week intervention.

III Present study

As mentioned above, the present study was conducted to investigate how undergraduate students' language mindsets and academic writing competence change after taking a one-semester EAP writing course by employing both variable- and person-centered approaches. Accordingly, the following research questions guided the present study.

- Research question 1: To what extent does a one-semester college EAP course change learners' language mindsets and English writing competence?
- Research question 2: What profiles exist among the participants with regard to language mindsets before and after the course, and how does the profile membership change over the semester (i.e. participants' movements across the profiles)?

IV Methods

1 Participants and instructional context

The participants in this study consisted of 319 undergraduate students (age 18–19 years) enrolled in a 10-week academic English writing course, predominantly designed for English as a second language (ESL) students, during their regular academic year at a large public university on the West Coast of the U.S. For admission to the university, they had met the score requirements on the TOEFL iBT by obtaining a minimum score of 80, which corresponds to the CEFR B2 or upper intermediate level (ETS, 2022a). Approximately 93% of them reported that they spoke Mandarin Chinese as a primary language, whereas the remaining 7% spoke Vietnamese, Korean, Spanish, and other languages. The majority (73%) had lived in the U.S. for less than 2 years. Regarding their majors, 50% were majoring in Engineering, Physics, Chemistry, and Biological Sciences, 35% in Business and Economics, and the remainder in Humanities, Social Sciences, Studio Art, and Film and Media Studies. As freshmen, they had entered the university with an average high school GPA (grade point average) of 3.97 (out of 4.4) or 90.2%.[1]

The academic English program at this college offered a number of courses and services to lower-division university students (freshman and sophomore levels), including a writing course (the focus of the present study), a reading and vocabulary development course, and a grammar course. This writing course was designed for students who were identified as

needing additional coursework for their academic English based on the placement test results. Students were required to complete this writing course, which was a pass/no pass course, to fulfill one of their lower-division requirements prior to their third year of study to ensure their academic success in the university. Most of the instructors for these courses had acquired English as an additional language. They had Ph.D. or M.A. degrees in applied linguistics. Many of them had taught or were teaching composition, with about 30% being part-time employees.

2 Intervention and procedure

The academic English writing course introduced in the present study shares a common objective with courses in other universities in the U.S., which is to improve students' academic writing competence. However, it differs from a regular academic English writing course in its curriculum and assessments. That is, the book *Mindset: The new psychology of success*, by Dweck (2006), provided the main theme and reading material for this course. The course packet for the current study consisted of a copy of the syllabus, course schedule, assignments, and two modules for classroom activities, focused on Dweck's (2006) notion of mindset. In addition, two reference books for grammar and writing components, such as Lane and Lange (2011) and Scarcella (2015), which were normally provided in the program's writing courses, were also adopted. Thus, this course's aim was two-fold: (1) to improve undergraduate students' competence in academic English writing, and (2) to change their mindset contextualized in academic English intervention from fixed to growth.

The target writing course required the registered students to complete:

- two in-class timed argumentative writing tests, at the beginning of the course (i.e. Time 1), and at the end of the course (i.e. Time 2);
- two writing assignments (after each module);
- instructor-assigned work, including short writes, in-class writing, journals, reading comprehension questions, and presentations;
- language development activities (e.g. attending writing workshops, visiting writing tutors, and participating in conversation groups); and
- eight quizzes on English writing (four quizzes per module).

After taking the aforementioned writing test on the first day of the course, the participants began the first module, titled 'How do people respond to challenges and failure?' based on the first two chapters of Dweck (2006): 'Chapter 1: The mindset', and 'Chapter 2: Inside the mindset'. After the first half of the semester, the students began the second module, titled 'Why is having growth mindsets important in achieving excellence and success?' based on the

last three chapters of Dweck (2006): ‘Chapter 3: The truth about ability and accomplishment’, ‘Chapter 4: Sports – The mindset of a champion’, and ‘Chapter 5: Business – Mindsets and leadership’.

Each module formed a core part of the learning materials, and each comprised the following four parts: (1) pre-reading vocabulary and discussion, (2) reading comprehension exercises, (3) vocabulary activities, and (4) oral language activities. The participants could thus preview target vocabulary before reading the assigned chapters, check their comprehension after the reading, review the target vocabulary exposed during the reading, and discuss their understanding of the content. Next, they were asked to write an argumentative essay based on their reflections on and evidence in the assigned chapters of Dweck (2006). Until the deadline (the first module was due in Week 5, the second in Week 10), the participants had to submit at least four drafts, which entailed instructors’ systematic feedback, including content feedback on the earlier drafts of papers and both rhetorical and language feedback on subsequent drafts.

The specific elements of feedback largely corresponded to the aforementioned target academic writing. First, regarding content and organization, the feedback focused on whether students’ writing:

- included adequate introductory information about the topic;
- included a clear thesis that responded to the prompt appropriately;
- organized and developed the topic/argument logically and thoughtfully;
- demonstrated understanding of the class text, connecting it to the topic;
- made sufficient use of the class text for support;
- reflected original ideas and critical thinking;
- included a conclusion that engaged the readers;
- demonstrated progress through the drafting process; and
- adhered to the assignment lengths for all drafts (2–3 pages/650 words).

Second, the feedback also provided information about the presentation and language elements of the students’ writing, determining if students’ writing:

- used tenses correctly;
- used specific verbs such as *claims*, *argues*, and *suggests* as well as appropriate vocabulary;
- varied sentence structure effectively and used cohesive devices such as pronoun reference, transitions, and parallelism appropriately;
- was well edited for grammar: few basic errors with nouns, verbs, pronouns, and prepositions; and

- followed punctuation and English writing conventions, including correct use of quotation marks.

3 Measurements

Three types of measurement were employed in this study: (1) mindset survey, (2) writing tests, and (3) course grade. First, to measure students' mindsets in academic English contexts, we adapted the LMI (Lou & Noels, 2017), which consists of 18 questions on a 6-point Likert scale, by revising the word 'language' in the original questionnaire to 'academic English'. In accordance with Lou et al. (2022), we computed indices by combining growth and fixed aspects, with the latter being reverse coded, so that the combined indices can present learners' growth (vs. fixed) mindset status. To guarantee students' comprehension of the questionnaire items, we included information on jargon, such as language intelligence (capacity to learn a language), biological factors (brain structures and hearing), and fluency (speaking effortlessly and with little hesitation). The inventory was given online in Weeks 1 (Time 1) and 10 (Time 2), and the reliabilities of the six composite items (GLB, L2B, and ASB items at Time 1 and Time 2) were 0.81, 0.78, 0.80, 0.76, 0.76, and 0.77, respectively.

Second, in Weeks 1 and 10 of the course, the participants were given a 50-minute writing test in their classrooms. The writing prompt was excerpted from Dweck (2006) and provided them with 8 to 10 example statements about mindsets (e.g. 'Your intelligence is something very basic about you that you can't change very much'; 'You can learn new things, but you can't really change how intelligent you are'; 'No matter how much intelligence you have, you can always change it quite a bit'; 'You can always substantially change how intelligent you are') and asked them whether they agreed or disagreed with the statement they selected. They were then asked to write an argumentative essay, using reasons and specific examples to support their opinions. The instructors scored the writing exams, blinding students' names and student identification numbers, and they were trained to follow TOEFL iBT's independent writing rubrics (ETS, 2022b) for scoring standards. The score was given holistically, ranging from 0 to 5, depending on the essay's discussion of the topic, organization, coherence, and correct use of grammar and vocabulary. Each writing piece was scored by two instructors, and one additional rater was invited when the score gap between the two raters was greater than 1. The interrater reliability was acceptable according to Pearson's Correlation Coefficients (r = .87; acceptable when > .70), Cronbach's alpha (α = .99, acceptable when > .90), Cohen's Kappa statistics (k = .81, acceptable when > .80), and Intra-class Correlation Coefficients (ICC; r = .89, acceptable when > .80).

As another measurement, the course grade was used to assess students' overall achievement in their academic English writing. The following comprised their course grade: two writing assignments after each module (60%), instructor-assigned work (10%), language development activities (5%), quizzes (5%), and in-class timed writing tests (20%). Students' writing assignments were scored by instructors based on the scoring standards with TOEFL iBT writing rubrics (ETS, 2022b).

4 Data analysis

Before conducting the statistical analyses, the dataset was carefully examined, and the cases with the missing values in one or more of our measurements were removed. Further, cases that responded to the LMI with the same response consistently (e.g. marking all the questionnaire items with '1' on the Likert scale) were removed as well. As a result, 20 cases were removed from the originally recruited sample, leaving 299 as the valid number. We used STATA Version 16 for the data analyses and Mplus Version 8.7 for replication.

For research question 1, which concerned the variable-centered approach, we used paired *t*-tests to examine if there were significant changes in the participants' growth (vs. fixed) language mindset across three themes (GLB, L2B, and ASB) and writing competence over the course of the semester.

For research question 2, we employed the person-centered approach, using both a latent profile analysis (LPA) and a latent transition analysis (LTA; an extended version of LPA). This approach with such statistical methods is gaining attention in the field of applied linguistics and SLA (e.g. Liu & Oga-Baldwin, 2022; Lou et al., 2022). Unlike bottom-up approaches, such as a cluster analysis, LPA and its variants are top-down approaches based on probabilistic models to provide statistical model fit indices (Hagenaars & McCutcheon, 2002). As our dataset was longitudinal, we additionally conducted LTA, also known as longitudinal latent profile analysis (LLPA), utilizing the following three steps according to Demaray et al.'s (2021) guidelines.

In Step 1, we first conducted two cross-sectional LPA models for each timepoint to select the best fitting models. Each LPA model was computed based on learners' growth (vs. fixed) mindset indices across three themes (GLB, L2B, and ASB) at each timepoint. To select best fitting models for each timepoint, we used a total of six criteria, including:

- the profile size criterion, i.e. each profile consists of more than 5% of the total sample (Celeux & Soromenho, 1996);
- three fit indices, i.e. Akaike information criterion (AIC) and Bayesian information criterion (BIC), with lower values indicating better fit, and log-likelihood value, with

higher values indicating better fit (Nylund et al., 2007);

- entropy value, i.e. an estimate of how identified profiles are distinctive from each other; > .80 for acceptable fit (Celeux & Soromenho, 1996); and
- Lo–Mendell–Rubin likelihood ratio (LMR) test, i.e. a significant *p*-value indicates that this profile fits better than the previous profile (Lo et al., 2001).

In Step 2, after identifying the best fitting models for each timepoint, we determined the optimum number of latent profiles for each timepoint and labeled each profile according to their statistical and theoretical characteristics. As the numbers of the best fitting LPA models at different timepoints could differ, the range of potential candidates for each timepoint should be explored to identify meaningful latent profiles (Demaray et al., 2021).

In Step 3, a total of six variables (i.e. GLB, L2B, and ASB each at Time 1 and Time 2) were used to build an LTA model including the determined numbers of latent profiles for each timepoint. After the computation, inferential statistics were conducted to further understand the LTA results.

V Results

1 Descriptive statistics and correlations

This section examines the descriptive statistics of the target variables, and the correlations between them. Table 1 presents preliminary analysis that includes means, standard deviation, skewness, and kurtosis. Language mindset indices ranged between 0.80 and 1.21 at Time 1, and between 1.06 and 1.39 at Time 2. Each language mindset index at the two time-points passed the normality tests for skewness and kurtosis. Further, we conducted a confirmatory factor analysis to examine the construct validity of the LMI items. The results of this analysis indicated that each of six LMI scales had an overall acceptable model fit. Table 2 represents the results of bivariate correlations between the target variables.

Table 1. Descriptive statistics of the target variables and validity of Language Mindsets Inventory (LMI) scale (n = 299).

Language mindsets index	*M* (*SD*)	Skewness		Kurtosis		Validity of LMI scale (CFA)				
		Value	Test	Value	Test	Chi-square	RMSEA	CFI	TLI	SRMR
GLB at Time 1	0.80 (1.64)	.32	$p > .05$	2.83	$p > .05$	$\chi^2/df = 0.88$, $p = .50$	.00	.99	.99	.02
L2B at Time 1	1.21 (1.38)	.42	$p > .001$	2.53	$p > .01$	$\chi^2/df = 1.24$, $p = .30$	.03	.99	.99	.03
ASB at Time 1	0.96 (1.54)	.02	$p > .05$	2.57	$p > .05$	$\chi^2/df = 0.19$, $p = .03$	.07	.99	.96	.04
GLB at Time 2	1.39 (1.81)	.14	$p > .05$	2.74	$p > .05$	$\chi^2/df = 2.44$, $p = .02$	.07	.99	.98	.03
L2B at Time 2	1.35 (1.72)	.24	$p > .05$	2.59	$p > .05$	$\chi^2/df = 4.69$, $p = .01$	.11	.98	.94	.06
ASB at Time 2	1.06 (1.61)	.46	$p > .001$	2.70	$p > .05$	$\chi^2/df = 2.77$, $p = .01$	.07	.99	.98	.02
Writing at Time 1	3.23 (0.86)	−.06	$p > .05$	2.30	$p > .001$					
Writing at Time 2	3.84 (0.89)	−.38	$p > .001$	3.42	$p > .05$					

Notes. GLB = general language intelligence beliefs; L2B = second language aptitude beliefs; ASB = age-sensitivity beliefs. The standards for goodness of fit for the CFA analyses, the following rules of thumbs (Jin & Lee, 2022; Lee et al., 2020, 2022) were endorsed: Chi-square test: $\chi^2/df < 5$. RMSEA < .08. CFI > .90. TLI > .90. SRMR < .08.

Table 2. Bivariate correlation between language mindset indices and writing test scores (n = 299).

Language mindsets index	1	2	3	4	5	6	7	8
1. GLB at Time 1	–							
2. L2B at Time 1	.53***	–						
3. ASB at Time 1	.29***	.51***	–					
4. GLB at Time 2	.33***	.44***	.29***	–				
5. L2B at Time 2	.28***	.42***	.28***	.67***	–			
6. ASB at Time 2	.18**	.34***	.45***	.60***	.69***	–		
7. Writing test scores at Time 1	−.01	.04	.03	.13*	.11	.17**	–	
8. Writing test scores at Time 2	.05	.10	−.01	.14*	.18**	.17**	.53***	–

Notes. $^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

2 Changes in language mindsets and English writing competence through one-semester EAP course

Research question 1 concerned the changes in the participants' growth (vs. fixed) language mindset and their English academic writing, as a result of a one-semester EAP writing course. Table 3 presents the results of the paired *t*-tests on these target variables. As can be seen in Table 3, the participants had higher levels of growth language mindsets than fixed language mindsets after taking the course (i.e. growth (vs. fixed) mindset indices > 0). The results of the paired *t*-tests revealed that the participants' growth (vs. fixed) language mindsets increased after taking the one-semester EAP writing course, though only the GLB showed a statistically significant small-sized improvement. Regarding English writing, the participants' scores increased significantly as a result of the one-semester EAP writing course with a medium effect size (d = 0.68), confirming its impact on improving students' academic writing competence.

Table 3. Results of paired *t*-tests (n = 299).

Target variable	Time 1 *M* (*SD*)	Time 2 *M* (*SD*)	*t*-value	Cohen's unbiased *d* (Hedges' *g*)
GLB	0.80 (1.64)	1.39 (1.81)	5.06***	0.34
L2B	1.21 (1.38)	1.35 (1.72)	1.43	0.09
ASB	0.96 (1.54)	1.06 (1.61)	1.08	0.07
Writing test scores	3.23 (0.86)	3.84 (0.89)	11.22***	0.68

Notes. GLB = general language intelligence beliefs; L2B = second language aptitude beliefs; ASB = age-sensitivity beliefs. Interpretations of Cohen's *d* for language mindset refer to Cohen's suggested benchmarks, where 0.2, 0.5, and 0.8 are considered small, medium, and large, respectively, whereas those for L2 writing refer to Lee and Lee's (2022) domain-specific benchmarks, where 0.3, 0.5, and 0.7 are considered small, medium, and large, respectively. $^{***}p < .001$.

3 Unearthing learner profiles over time

For research question 2, we employed the person-centered analytical approach. We first conducted a latent profile analysis (LPA) to unearth learner profiles before and after the EAP writing program, if any, and then employed latent transition analysis (LTA) to explore how profile membership changed over time.

Tables 4 and 5 provide fit indices for the 2- through 5-profile solutions for Time 1 and Time 2, respectively, and we selected 3-profile models for both timepoints. For Time 1, the 4- and 5-profile solutions had profiles with less than 15 samples, violating the 5% rule. When compared with the 2-profile solution, the 3-profile solution had higher LL, and lower AIC and BIC values. For Time 2, a similar approach was applied. The 4- and 5-profile solutions involved a profile with less than 21 students. The 3-profile solutions had better fit indices than the 2-profile solution. Overall, for both timepoints we selected the most parsimonious solution (i.e. the 3-profile solution) considering the interpretability and parsimony of different solutions (Nylund et al., 2007).

Table 4. Fit indices for L2 mindset-based profile solutions at Time 1: One latent profile analysis (LPA) cross-sectional model.

Number of profiles	Profile size	LL	AIC	BIC	Entropy	LMR test
2	190, 109	−1,539.83	3,099.65	3,136.65	.95	> .001
3*	181, 45, 73	−1,525.00	3,078.00	3,129.81	.92	> .001
4	168, 14, 53, 64	−1,509.64	3,055.29	3,121.89	.92	> .001
5	7, 165, 14, 48, 65	−1,497.79	3,039.57	3,120.98	.93	> .01

Notes. * The selected best model. Underlined values for the profile size indicate profiles consisting of less than or equal to 5% (i.e. 15) of the total sample. LL = log-likelihood. AIC = Akaike information criterion. BIC = Bayesian information criterion. LMR test = Lo–Mendell–Rubin adjusted likelihood ratio test.

Table 5. Fit indices for L2 mindset-based profile solutions at Time 2: One latent profile analysis (LPA) cross-sectional model.

Number of profiles	Profile size	LL	AIC	BIC	Entropy	LMR test
2	194, 105	−1,570.95	3,161.90	3,198.91	.96	> .001
3*	157, 100, 42	−1,533.81	3,095.63	3,147.43	.94	> .001
4	5, 156, 93, 45	−1,514.86	3,065.71	3,132.32	.95	> .001
5	5, 146, 14, 92, 42	−1,500.28	3,044.56	3,125.97	.95	> .001

Notes. * The selected best model. Underlined values for the profile size indicate profiles consisting of less than or equal to 5% (i.e. 15) of the total sample. LL = log-likelihood. AIC = Akaike information criterion. BIC = Bayesian information criterion. LMR test = Lo–Mendell–Rubin adjusted likelihood ratio test.

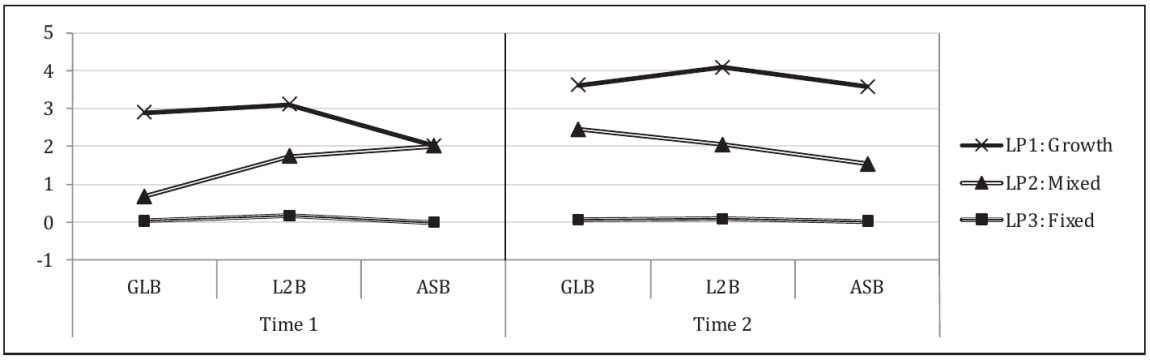


Figure 1. Growth (vs. fixed) language mindsets for latent profiles at Time 1 and Time 2 (one latent-transition-analysis longitudinal model).
Note. LP = latent profile.

Based on the results of the two LPA models, we employed an LTA approach by including six variables (i.e. GLB, L2B, and ASB each for Time 1 and Time 2) into a single statistical model to more accurately investigate how learners changed their growth (vs. fixed) language mindset profiles over time. The results indicated that the LTA model had LL, AIC, and BIC values of −3,019.538, 6,103.08, and 6,221.49, respectively, along with the entropy values of 0.80. Figure 1 illustrates the means of language mindset indices for each profile at both time-points. The left side of the figure is for three profiles at Time 1 and the right for those at Time 2; both sides had similar patterned profiles.

First, the profiles on top (single line) had larger growth (vs. fixed) language mindset scores across GLB, L2B, and ASB themes, and were labeled ‘Growth profile’. Second, the profiles in the middle (double line) had smaller growth (vs. fixed) language mindset scores than the first profiles, and were labeled ‘Mixed profile’. Third, the profiles at the bottom (triple line) had the smallest growth (vs. fixed) language mindset scores; accordingly, we labeled them ‘Fixed profile’.

More specifically, Table 6 reports descriptive statistics of growth (vs. fixed) language mindset values for each latent profile at both timepoints from the one LTA longitudinal model. For Time 1, ANOVA results indicated that 1LP1 had the largest growth (vs. fixed) language mindset values, followed by 1LP2 and 1LP3, and the language mindset gap between these profiles were significantly different ($p < .05$; except for the ASB). For Time 2, the identified profiles followed the same patterns as those at Time 1; 2LP1 had the largest language mindset values followed by 2LP2 and 2LP3 ($p < .05$).

Table 6. Descriptive statistics of growth (vs. fixed) language mindset for each latent profile at Time 1 and Time 2 (one LTA longitudinal model).

Time 1							**Time 2**						
LP	**GLB**		**L2B**		**ASB**		**LP**	**GLB**		**L2B**		**ASB**	
1LP1: Growth (*n* = 63)	2.89 (1.15)	G: 4.98 F: 2.10	3.11 (0.79)	G: 5.41 F: 2.30	2.02 (1.41)	G: 4.99 F: 2.98	2LP1: Growth (*n* = 46)	3.62 (1.42)	G: 5.23 F: 1.62	4.09 (0.64)	G: 5.71 F: 1.62	3.58 (0.84)	G: 5.72 F: 2.14
1LP2: Mixed (*n* = 80)	0.68 (1.31)	G: 3.90 F: 3.22	1.74 (0.61)	G: 4.81 F: 3.01	2.00 (1.24)	G: 4.82 F: 2.81	2LP2: Mixed (*n* = 98)	2.45 (1.11)	G: 4.61 F: 2.16	2.05 (0.95)	G: 4.78 F: 2.73	1.53 (1.15)	G: 4.62 F: 3.01
1LP3: Fixed (*n* = 156)	0.03 (1.19)	G: 3.60 F: 3.57	0.17 (0.73)	G: 3.89 F: 3.72	−0.01 (1.01)	G: 3.70 F: 3.71	2LP3: Fixed (*n* = 155)	0.06 (0.99)	G: 3.40 F: 3.33	0.09 (0.95)	G: 3.63 F: 3.54	0.01 (0.90)	G: 3.52 F: 3.51
ANOVA (α = 5%)	1LP1 > 1LP2 > 1LP3		1LP1 > 1LP2 > 1LP3		1LP1 ≈ 1LP2 > 1LP3		ANOVA (α = 5%)	2LP1 > 2LP2 > 2LP3		2LP1 > 2LP2 > 2LP3		2LP1 > 2LP2 > 2LP3	

Notes. Values inside and outside the parentheses are standard deviations and means, respectively. GLB = general language intelligence beliefs; L2B = second language aptitude beliefs; ASB = age-sensitivity beliefs; LP = latent profile; ANOVA = analysis of variance; G = the average of the participants' responses on the items related to the growth mindset; F = the average of the participants' responses on the items related to the fixed mindset.

Table 7. Academic achievement according to latent profiles at Time 1 and Time 2: One latent transition analysis (LTA) longitudinal model.

Time 1		**Time 2**		
LP	**Writing test score**	**LP**	**Writing test score**	**Course grade**
1LP1: Growth (*n* = 63)	3.39 (0.79)	2LP1: Growth (*n* = 46)	4.13 (0.81)	85.28 (7.18)
1LP2: Mixed (*n* = 80)	3.13 (0.82)	2LP2: Mixed (*n* = 98)	3.89 (0.84)	83.59 (7.01)
1LP3: Fixed (*n* = 156)	3.13 (0.78)	2LP3: Fixed (*n* = 155)	3.66 (0.86)	81.57 (7.40)
ANOVA (α = 5%)	1LP1 ≈ 1LP2 ≈ 1LP3	ANOVA (α = 5%)	2LP1 ≈ 2LP2 > 2LP3	2LP1 ≈ 2LP2 > 2LP3

Notes. The values in parentheses are standard errors. Outside the parentheses are standard deviations and means, respectively. LP = latent profile. ANOVA = analysis of variance.

Table 7 represents the results of a follow-up analysis to further check how these profiles differed in terms of their writing test scores at both time-points and their course grades. For Time 1, we found that learners across different profiles at Time 1 did not differ from each other. However, for Time 2, the results showed that the Growth and Mixed profile students had higher writing scores and course grades than Fixed profile students.

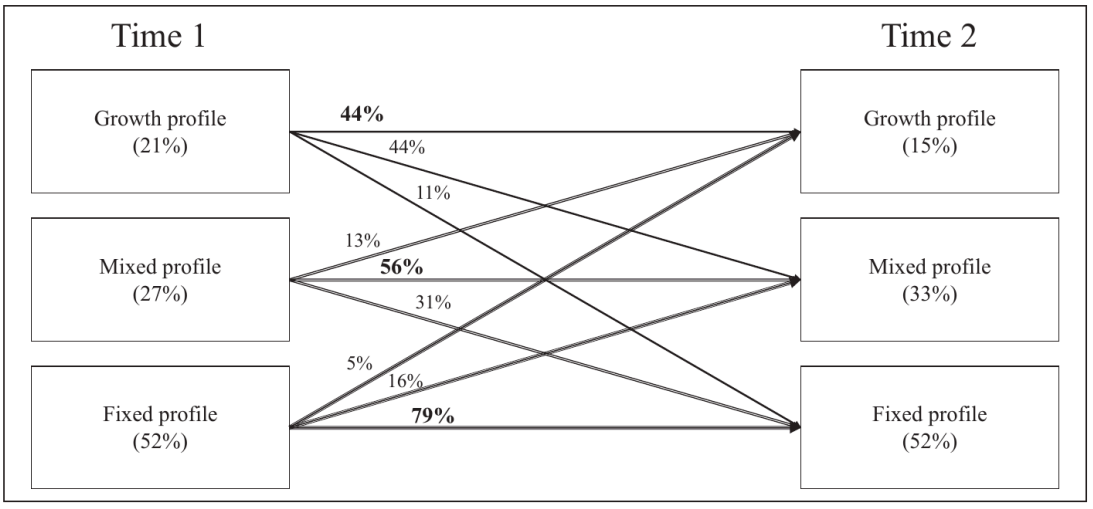


Figure 2. Latent profile transition patterns (one latent-transition-analysis longitudinal model).

Last but not least, as we could identify how participants changed their profiles from Time 1 to Time 2 by using the LTA approach, we additionally checked the transition patterns of profile membership among the participants. Figure 2 illustrates the profile transition patterns. First, among students (*n* = 63) with a Growth profile at Time 1, about 44% maintained it till Time 2, another 44% moved to Mixed profile with lower growth (vs. fixed) language mindset scores, and the remaining 11% experienced a substantial decrease in their growth (vs. fixed) language mindset status, becoming Fixed profile students. Second, students with a Mixed profile at

Time 1 (n = 80) showed different transition patterns; about 13% increased their growth (vs. fixed) language mindset scores and had a Growth profile at Time 2, 56% maintained their profile status, and 31% changed their profile to Fixed profile with decreased growth (vs. fixed) language mindset scores at Time 2. Third, students with a Fixed profile at Time 1 (n = 156) showed the most stable transition patterns when compared with their peers. About 79% of them maintained their profile status, whereas only 5% and 16% became Growth profile and Mixed profile type students, respectively, at Time 2 by having increased growth (vs. fixed) language mindset scores.

VI Discussion

The present study was conducted to provide empirical evidence regarding the value of the EAP course based on the mindset theory (Dweck, 2006) for the changes in undergraduate students' language mindset and academic English writing competence. This section will present our discussion on the results of the study as well as its limitations.

1 The impact of an EAP course based on the mindset theory

Regarding research question 1, the findings of the present study showed that an EAP writing course was beneficial in improving undergraduate students' academic English writing competence, which concurs with previous studies on EAP writing programs (e.g. Crossley et al., 2016; Park, 2015). Although it remains unexplored which component of our target EAP course contributed to the development of academic English writing, the observed benefits could have stemmed from the combined effect of instructors' feedback on students' drafts (e.g. Bitchener & Ferris, 2012; Ferris, 2002; Leki, 2006), exposure to English academic vocabulary (Coxhead, 2016), and genre-based instruction (Hyland, 2007, 2008), to list a few. In addition, the effect size related to the improvement in writing (d = 0.68) could be considered medium, in view of the domain-specific effect size benchmarks proposed in a recently published meta-analysis of the impact of EAP writing programs (Lee & Lee, 2022), where 0.3, 0.5, and 0.7 are considered small, medium, and large, respectively. Meanwhile, this result should be read with some caution, due to the lack of a comparison (or control) group in the present study (for more on this point, see Section VI.3). Regarding the mindsets, the participants' growth (vs. fixed) language mindset increased in terms of GLB, L2B, and ASB, but it was only in the first dimension (i.e. GLB) that they showed a significant increase. One possible explanation for a relatively larger increase in GLB could be that the mindset intervention was largely based on Dweck's (2006) research, which discusses the mindset

system in general domains; thus, this intervention may have been more sensitive to the increase in GLB (i.e. general language intelligence beliefs) than L2B and ASB, which constitute L2-specific issues.

Some pedagogical suggestions can be made based on these findings. First, like the EAP writing course introduced in this study, EAP instructors in other contexts may adopt a similar instructional approach, which would effectively serve a dual purpose of providing English writing instruction and enhancing growth mindsets. In view of Lou and Noels (2020), students in such courses can be asked to 'read and write a reflection on an article about the incremental view of intelligence' or read other articles related to the value of growth mindsets (p. 3). Second, in conducting the EAP writing course based on the mindset theory, instructors are encouraged to ensure that their learners fully acknowledge and internalize the concept of growth language mindsets throughout the course, so that learners can maintain positive attitudes and resilience towards developing their L2 writing competence. This suggestion aligns with the findings of Yao, Wang, et al. (2021) that growth mindsets may play a significant role in sustaining L2 writing motivation. For these reasons, we believe English writing lessons (EAP in this context) based on the mindset theory has significant potential.

2 Changes in profile membership in terms of language mindsets as a result of a one-semester course

The purpose of employing the person-centered analysis was to supplement the overall findings related to the whole group (i.e. entire sample) observed in the variable-centered analysis. With such an approach being used to identify learning patterns of a subset of the participants, our model revealed the three-group solution (i.e. Growth profile, Mixed profile, and Fixed profile), which echoes the findings of Lou et al. (2022) and Yao, Wang, et al. (2021). It is noteworthy, however, that the characteristics of our three profiles may be slightly different from those identified in previous studies. Accordingly, these characteristics should be interpreted in relative, rather than absolute, terms.

Having identified three learner profiles in terms of the language mindset, we set out to examine the differences among them in academic writing competence at Time 1 and Time 2. Whereas no significant difference was found among the three identified profiles at Time 1, Growth and Mixed profiles performed significantly better than the Fixed profile after the one-semester course, indicating the positive role of growth mindsets in academic learning (Blackwell et al., 2007; Yeager et al., 2016). This finding may be best accounted for by the proposition that those with growth mindsets, unlike those with fixed mindsets, may employ mastery-oriented strategies as opposed to helpless-oriented ones (e.g. Burnette et al., 2013; Lou & Noels, 2019a) or be more engaged in their learning (Lou et al., 2022). Another finding

that the Growth profile group achieved significantly better course grades than the Fixed profile group further corroborates the important role of the growth mindset, through which one may have set a clearer goal (i.e. developing one's academic writing and having a high grade in our context) and made greater efforts to drive themselves toward achieving this goal throughout the semester.

Finally, the result related to the transition patterns of profile membership, as revealed through the implementation of the LTA longitudinal model, deserves some discussion. Addressing Lou et al.'s (2022) call for a longitudinal study on language mindsets, the overall finding in this regard showed that dynamic shifts in the profile membership could occur through mindset intervention (see Figure 2). A closer examination of the transition patterns revealed that about 55% of the Growth profile group shifted to either the Mixed or Fixed profile, and 31% of the Mixed profile shifted to the Fixed profile; thus, those in these profiles could be suggested to have shown some decrease in their growth mindsets after the one-semester intervention. One possible explanation for this finding comes from Lou and Noels (2019b), who claim that growth mindsets may have some adverse effects on learners when they face a new task, by making them become more self-critical. The task of fostering one's English academic writing competence may have been perceived as a challenging new task. It may have also contributed to a decrease in growth language mindset. Similarly, as found in Yao, Wang, et al. (2021), an intervention may bring about diverse perceptions thereof, which would result in dynamic shifts of the profile membership. Such shifts may have also risen from the interaction between learners' competence at the time of the intervention (Burnette et al., 2013; Dweck & Leggett, 1988), classroom-related variables (e.g. school norm; Yeager et al., 2019), and the target mindset intervention, which should be explored in future research on mindsets.

Overall, the person-centered approach has enabled us to suggest that the effects of mindset intervention in the EAP context may be differential among the target students. In this regard, we concur with Lou and Noels (2019b) that 'scholars and teachers should take a student-centered approach to understand how to customize mindset interventions that are effective for different language learners' (p. 8). Our approach in this study may offer some evidence in support of this assertion.

As for pedagogical suggestions, EAP writing instructors are encouraged to pay attention to those with fixed mindsets at the beginning of the course, as they are more likely to struggle with the change to a growth mindset. These learners may benefit from personal encouragement or information on the benefits of English writing programs, both of which may positively influence their growth mindsets. At the same time, instructors are recommended to keep their eyes on learners with growth mindsets at the beginning of the course, as some may experience unforeseen challenges in L2 writing and change their language mindsets to fixed

ones. In conclusion, if instructors can give their students enough individual attention in a way that promote growth language mindsets as well as L2 writing competence, students with diverse language mindset statuses would likely benefit.

3 Limitations

The present study had some limitations, three of which are discussed here. First, for practical and ethical reasons, we could not sample a comparison group. Although a pre–post design has some methodological strengths, without a longitudinal comparison group, any observed changes via the mindset intervention in the present study should be interpreted with due caution. Second, the present study predominantly focused on language mindsets and English academic writing as the target variables, leaving out the interaction effects between these and other important learner variables such as anxiety, motivation, self-confidence, and grit. The investigation into the relationships between language mindsets and these variables has only started to emerge (e.g. Khajavy et al., 2021; Lou et al., 2022), and pursuing this effort with the experimental design would shed further light on the role of language mindsets in promoting language achievement. Last, we did not measure the teachers' mindsets (e.g. their beliefs about whether L2 learners' aptitude is malleable), although they may have some potential to influence learners' language mindsets (Lou & Noels, 2017). It would be worthwhile for future intervention research to measure teachers' general and language-specific mindsets and examine if teachers' mindsets may influence learners' mindsets and their language learning.

VII Conclusions

The present study set out to examine the effects of a 10-week EAP writing course equipped with the materials of mindset intervention on undergraduate students' language mindsets and academic writing competence. With this research goal, it was found that the students' mindset could be positively influenced by the mindset intervention, and their writing could also improve to a moderate extent. By employing a person-centered approach, the present study further revealed that there were distinctive learner profiles in our dataset in terms of their language mindsets, and that each may show different learning trajectories even when exposed to the same instructional environment.

The findings of this study appear to cast a new light on the future directions of research on language mindsets in two significant ways. First, it is one of the first longitudinal studies in SLA to have successfully implemented a language learning course equipped with a mindset intervention and demonstrated its impact on both developing English skills and changing the level of language mindsets (i.e. GLB in our study). Researchers may take these findings into consideration and aim to further explore the effects of this type of intervention on other

linguistic dimensions and skill-specific mindsets (e.g. Khajavy et al., 2022; Waller & Papi, 2017). Second, we have also demonstrated the value of the simultaneous execution of both variable-centered and person-centered analyses. Such an approach, if implemented along with the experimental design, may contribute to the development of more individualized instruction (for a similar viewpoint, see Lee et al., 2019) in terms of the target variables, and we believe that language mindset research will be a considerable beneficiary with this line of approach.

Acknowledgements

The authors would like to thank Dr. Tina Matuchniak, who is now the Director of the Writing Program at Long Beach State University, for collecting data which were used in this research.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work was supported by the Hwarangdae Research Institute at the Korea Military Academy (22-군학-01).

ORCID iDs

Hansol Lee https://orcid.org/0000-0002-6912-7128

Jang Ho Lee https://orcid.org/0000-0003-2767-3881

Note

1. This university calculates the high school GPA of their applicants based on a ‘weighted-capped’ scale with a maximum of 4.4 points, by assigning an extra grade point for achievements such as honors or college-level courses taken in high school.

References

Birdsong, D. (1999). Introduction: Whys and why nots of the critical period hypothesis for second language acquisition. In Birdsong, D. (Ed.), *Second language acquisition and the critical period hypothesis* (pp. 1–22). Lawrence Erlbaum.

Bitchener, J., & Ferris, D.R. (2012). *Written corrective feedback in second language acquisition and writing*. Routledge.

Blackwell, L.S., Trzesniewski, K.H., & Dweck, C.S. (2007). Implicit theories of intelligence predict achievement across an adolescent transition: A longitudinal study and an intervention. *Child Development, 78*, 246–263.

Burnette, J.L., O'Boyle, E.H., VanEpps, E.M., Pollack, J.M., & Finkel, E.J. (2013). Mind-sets matter: A meta-analytic review of implicit theories and self-regulation. *Psychological Bulletin, 139*, 655–701.

Carroll, J.B. (1981). Twenty-five years of research on foreign language aptitude. In Diller, K.C. (Ed.), *Individual differences and universals in language learning aptitude* (pp. 83–118). Newbury House.

Celeux, G., & Soromenho, G. (1996). An entropy criterion for assessing the number of clusters in a mixture model. *Journal of Classification, 13*, 195–212.

Chen, J.A., & Tutwiler, M.S. (2017). Implicit theories of ability and self-efficacy: Testing alternative social cognitive models to science motivation. *Zeitschrift für Psychologie*, *225*, 127–136.

Coxhead, A. (2016). Acquiring academic and disciplinary vocabulary. In Hyland, K., & P. Shaw (Eds.), *The Routledge handbook of English for academic purposes* (pp. 177–190). Routledge.

Crossley, S.A., Kyle, K., & McNamara, D.S. (2016). The development and use of cohesive devices in L2 writing and their relations to judgments of essay quality. *Journal of Second Language Writing, 32*, 1–16.

Crowther, D., Kim, S., Lee, J., Lim, J., & Loewen, S. (2021). Methodological synthesis of cluster analysis in second language research. *Language Learning, 71*, 99–130.

DeKeyser, R.M. (2000). The robustness of critical period effects in second language acquisition. *Studies in Second Language Acquisition, 22*, 499–533.

Demaray, M.K., Malecki, C.K., Ryoo, J.H., & Summers, K.H. (2021). Deconstructing bullying roles: A longitudinal latent profile analysis of bullying participant behaviors for students in grades 4 through 12. *Journal of School Psychology*, *86*, 32–48.

Dweck, C.S. (1999). *Self-theories: Their role in motivation, personality, and development*. Psychology Press.

Dweck, C.S. (2006). *Mindset: The new psychology of success*. Random House.

Dweck, C.S., & Leggett, E.L. (1988). A social-cognitive approach to motivation and personality. *Psychological Review, 95*, 256–273.

Dweck, C.S., & Yeager, D.S. (2019). Mindsets: A view from two eras. *Perspectives on Psychological Science, 14*, 481–496.

Dweck, C.S., Chiu, C.Y., & Hong, Y.Y. (1995). Implicit theories and their role in judgments and reactions: A word from two perspectives. *Psychological Inquiry, 6*, 267–285.

Educational Testing Service (ETS). (2022a). *Comparing TOEFL iBT scores*. Educational Testing Service. Available at: https://www.ets.org/toefl/institutions/scores/compare/ (accessed November 2022).

Educational Testing Service (ETS). (2022b). *TOEFL iBT: Independent writing rubrics* (Scoring standards). Educational Testing Service. Available at: https://www.ets.org/content/dam/ets-org/pdfs/toefl/toefl-ibt-writing-rubrics.pdf (accessed November 2022).

Eren, A., & Rakıcıoğlu-Söylemez, A. (2020). Language mindsets, perceived instrumentality, engagement and graded performance in English as a foreign language students. *Language Teaching Research*. Epub ahead of print 22 September 2020. DOI: 10.1177/1362168820958400

Ferris, D.R. (2002). *Treatment of error in second language student writing*. University of Michigan Press.

Gardner, H. (1993) *Multiple intelligences: The theory in practice*. Basic Books.

Hagenaars, J.A., & McCutcheon, A.L. (Eds.). (2002). *Applied latent class analysis*. Cambridge University Press.

Hong, Y.-y., Chiu, C.-y., Dweck, C.S., Lin, D.M.-S., & Wan, W. (1999). Implicit theories, attributions, and coping: A meaning system approach. *Journal of Personality and Social Psychology*, *77*, 588–599.

Horwitz, E.K. (1988). The beliefs about language learning of beginning university foreign language students. *The Modern Language Journal*, *72*, 283–294.

Howard, M.C., & Hoffman, M.E. (2018). Variable-centered, person-centered, and person-specific approaches: Where theory meets the method. *Organizational Research Methods*, *21*, 846–876.

Hyland, K. (2007). Genre pedagogy: Language, literacy and L2 writing instruction. *Journal of Second Language Writing*, *16*, 148–164.

Hyland, K. (2008). Genre and academic writing in the disciplines. *Language Teaching*, *41*, 543–562.

Hyland, K. (2016). General and specific EAP. In Hyland, K., & P. Shaw (Eds.), *The Routledge handbook of English for academic purposes* (pp. 17–29). Routledge.

Jin, S., & Lee, H. (2022). Willingness to communicate and its high-evidence factors: A meta-analytic structural equation modeling approach. *Journal of Language and Social Psychology*, *41*, 716–745.

Khajavy, G.H., MacIntyre, P.D., & Hariri, J. (2021). A closer look at grit and language mindset as predictors of foreign language achievement. *Studies in Second Language Acquisition*, *43*, 379–402.

Khajavy, G.H., Pourtahmasb, F., & Li, C. (2022). Examining the domain-specificity of language mindset: A case of L2 reading comprehension. *Innovation in Language Learning and Teaching, 16*, 208–220.

Lane, J., & Lange, E. (2011). *Writing clearly: Grammar for editing*. 3rd edition. Heinle.

Lee, H., & Lee, J.H. (2022). Interpreting the effectiveness of academic English writing programmes in higher education: A meta-analysis. *Oxford Review of Education*. Epub ahead of print 17 May 2022. DOI: 10.1080/03054985.2022.2072284

Lee, H., Jung, G., & Lee, J.H. (2022). Simple view of second language reading: A meta-analytic structural equation modeling approach. *Scientific Studies of Reading*, *26*, 585–603.

Lee, H., Warschauer, M., & Lee, J.H. (2019). Advancing CALL research via data-mining techniques: Unearthing hidden groups of learners in a corpus-based L2 vocabulary learning experiment. *ReCALL*, *31*, 135–149.

Lee, H., Warschauer, M., & Lee, J.H. (2020). Toward the establishment of a data-driven learning model: Role of learner factors in corpus-based second language vocabulary learning. *The Modern Language Journal*, *104*, 345–362.

Leki, I. (2006). ‘You cannot ignore’: L2 graduate students’ response to discipline-based written feedback. In Hyland, K., & F. Hyland (Eds.), *Feedback in second language writing* (pp. 266–285). Cambridge University Press.

Liu, M., & Oga-Baldwin, W.Q. (2022). Motivational profiles of learners of multiple foreign languages: A self-determination theory perspective. *System, 106*, 102762.

Lo, Y., Mendell, N.R., & Rubin, D.B. (2001). Testing the number of components in a normal mixture. *Biometrika, 88*, 767–778.

Lou, N.M., & Noels, K.A. (2016). Changing language mindsets: Implications for goal orientations and responses to failure in and outside the second language classroom. *Contemporary Educational Psychology, 46*, 22–33.

Lou, N.M., & Noels, K.A. (2017). Measuring language mindsets and modeling their relations with goal orientations and emotional and behavioral responses in failure situations. *The Modern Language Journal, 101*, 214–243.

Lou, N.M., & Noels, K.A. (2019a). Language mindsets, meaning-making, and motivation. In Lamb, M., Csizér, K., Henry, A., & S. Ryan (Eds.), *The Palgrave handbook of motivation for language learning* (pp. 537–559). Palgrave Macmillan.

Lou, N.M., & Noels, K.A. (2019b). Promoting growth in foreign and second language education: A research agenda for mindsets in language learning and teaching. *System, 86*, 102126.

Lou, N.M., & Noels, K.A. (2019c). Sensitivity to language-based rejection in intercultural communication: The role of language mindsets and implications for migrants' cross-cultural adaptation. *Applied Linguistics, 40*, 478–505.

Lou, N.M., & Noels, K.A. (2020). Breaking the vicious cycle of language anxiety: Growth language mindsets improve lower-competence ESL students' intercultural interactions. *Contemporary Educational Psychology, 61*, 101847.

Lou, N.M., Chaffee, K.E., & Noels, K.A. (2022). Growth, fixed and mixed mindsets: Mindset system profiles in foreign language learners and their roles in engagement and achievement. *Studies in Second Language Acquisition, 44*, 607–632.

Matsumoto, H., Nakayama, A., & Hiromori, T. (2013). Exploring the development of individual difference profiles in L2 reading. *System, 41*, 994–1005.

Mazgutova, D., & Kormos, J. (2015). Syntactic and lexical development in an intensive English for academic purposes programme. *Journal of Second Language Writing, 29*, 3–15.

Mercer, S., & Ryan, S. (2010). A mindset for EFL: Learners' beliefs about the role of natural talent. *ELT Journal, 64*, 436–444.

Nylund, K.L., Asparouhov, T., & Muthén, B.O. (2007). Deciding on the number of classes in latent class analysis and growth mixture modeling: A Monte Carlo simulation study. *Structural Equation Modeling, 14*, 535–569.

Park, J. (2015). Implementing extensive reading in university EAP writing classes. Unpublished doctoral dissertation, University of Hawai'i at Mānoa, HI, Honolulu.

Ryan, S., & Mercer, S. (2012). Implicit theories: Language learning mindsets. In Mercer, S., Ryan, S., & M. Williams (Eds.), *Psychology for language learning: Insights from research, theory and practice* (pp. 74–89). Palgrave Macmillan.

Scarcella, R.C. (Ed.). (2015). *Knowing language*. Fountainhead Press.

Skehan, P. (1996). A framework for the implementation of task-based instruction. *Applied Linguistics, 17*, 38–62.

Staples, S., & Biber, D. (2015). Cluster analysis. In Plonsky, L. (Ed.), *Advancing quantitative methods in second language research* (pp. 243–274). Routledge.

von Eye, A., & Bogat, G.A. (2006). Person-oriented and variable-oriented research: Concepts, results, and development. *Merrill–Palmer Quarterly*, *52*, 390–420. Available at: https://digitalcommons.wayne.edu/mpq/vol52/iss3/3/ (accessed November 2022).

Waller, L., & Papi, M. (2017). Motivation and feedback: How implicit theories of intelligence predict L2 writers' motivation and feedback orientation. *Journal of Second Language Writing*, *35*, 54–65.

Wang, H., Peng, A., & Patterson, M.M. (2021). The roles of class social climate, language mindset, and emotions in predicting willingness to communicate in a foreign language. *System*, *99*, 102529.

Yamamori, K., Isoda, T., Hiromori, T., & Oxford, R.L. (2003). Using cluster analysis to uncover L2 learner differences in strategy use, will to learn, and achievement over time. *International Review of Applied Linguistics*, *41*, 381–409.

Yao, Y., Guo, N.S., Li, C., & McCampbell, D. (2021). How university EFL writers' beliefs in writing ability impact their perceptions of peer assessment: Perspectives from implicit theories of intelligence. *Assessment and Evaluation in Higher Education*, *46*, 151–167.

Yao, Y., Wang, W., & Yang, X. (2021). Perceptions of the inclusion of automatic writing evaluation in peer assessment on EFL writers' language mindsets and motivation: A short-term longitudinal study. *Assessing Writing*, *50*, 100568.

Yeager, D.S., Hanselman, P., Walton, G.M., et al. (2019). A national experiment reveals where a growth mindset improves achievement. *Nature*, *573*, 364–369.

Yeager, D.S., Romero, C., Paunesku, D., et al. (2016). Using design thinking to improve psychological interventions: The case of the growth mindset during the transition to high school. *Journal of Educational Psychology*, *108*, 374–391.

Yu, J., & McLellan, R. (2020). Same mindset, different goals and motivational frameworks: Profiles of mindset-based meaning systems. *Contemporary Educational Psychology*, *62*, 101901.